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Analysis of Differences in WRF Simulations of East Asian Summer Monsoon in 2022 Based on Planetary Boundary Layer and Convective Parameterization Schemes

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1. Introduction

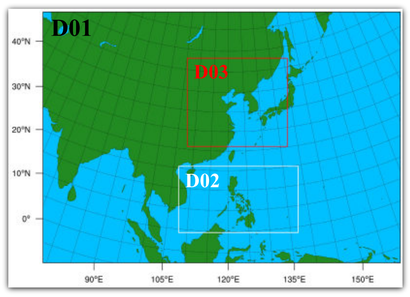
- In **East Asia (EA)**, about half of the annual rainfall occurs in the summer, making the prediction of the **East Asian Summer Monsoon (EASM)** particularly challenging due to various influencing factors.
- Understanding East Asian summer precipitation and improving simulation accuracy through optimal experimental design is essential. Summer rainfall in East Asia is primarily associated with **convective precipitation**.
- The **complex topography** of this region influences the exchange of heat, moisture, and momentum between land, ocean, and atmosphere, as well as boundary mixing processes.
- While the overall summer precipitation in East Asia in 2022 was average, the Korean Peninsula set a record for maximum hourly rainfall, and southeastern China saw its second-highest June to September precipitation since 1961.
- This study used the **Weather Research and Forecasting (WRF)** model to analyze 2022 summer precipitation in East Asia.
- For Cumulus Parameterization Schemes (CPS), we utilized the **KIM Simplified Arakawa-Schubert scheme (KSAS)** and the **Multi-scale Kain-Fritsch (MSKF)** schemes.
- By employing PBL methods such as the **Yonsei University (YSU)** and **Asymmetric Convective Model (ACM2)**, the study assessed their performance and impact on weather prediction by analyzing differences in precipitation amounts.

2. Data & Methods

Model configuration

Name	Advanced Research WRF (ARW) v4.4			
	01 YSU_KSAS	02 YSU_MSKF	03 ACM2_KSAS	04 ACM2_MSKF
Horizontal Resolution| (Domain)	12 km (823×570) / 4 km (826×451) / 4 km (691×601)			
Vertical Layers	35 hybrid levels			
Microphysics	WSM6			
Cumulus Parameterization	KSAS	MSKF	KSAS	MSKF
Radiation	RRTMG			
Planetary Boundary Layer	YSU	YSU	ACM2	ACM2
Land Surface	Noah LSM			
Initial & Boundary data	ERA5 (0.25° and 6 h)			
Analysis Period	0000 UTC on 1st June to 0000 UTC on 1st September 2022			

Model Domain



1) East Asia (CORDEX phase II Domain)
   - 1 – 52 °N / 81 –Z 155 °E
2) Low Latitude (Philippine Sea)
   - 6 – 18 °N / 109 – 136 °E
3) Mid Latitude (Korean Peninsula)
   - 28 – 46 °N / 110 – 136 °E

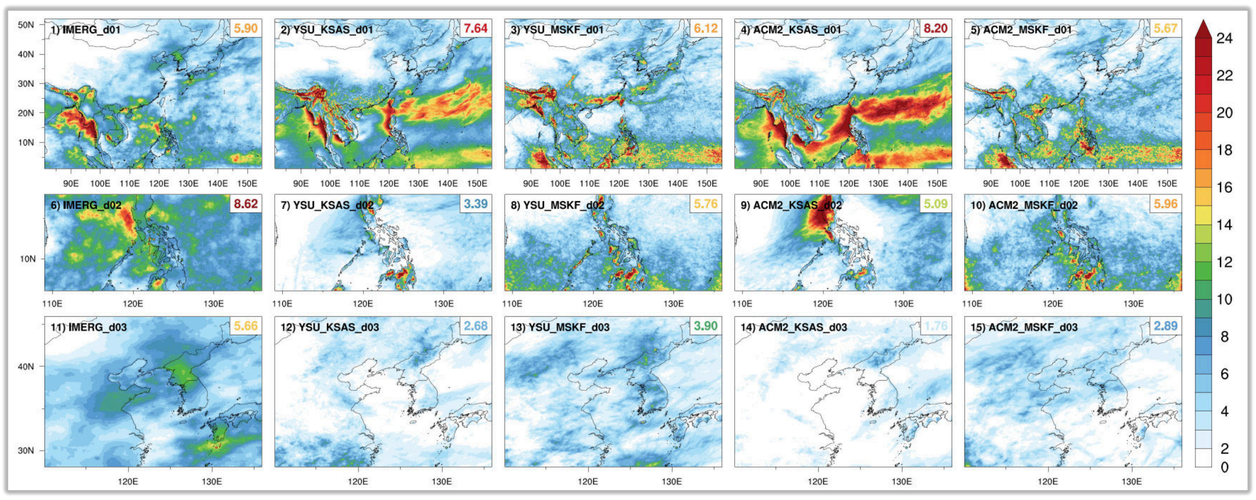
Data

- **IMERG (Integrated Multi-satellitE Retrievals for GPM)** data is used for precipitation validation (30 min, 0.1°)
- **WRF model output** has been bilinearly interpolated to 0.1° to compare with IMERG data (both 12 km and 4 km experiment)
- **ERA5 (ECMWF Reanalysis v5)** data is used for initial and boundary condition (6 hourly, 0.25°)

3. Results

East Asia 2022 Summer Mean Precipitation

Unit: mm/day



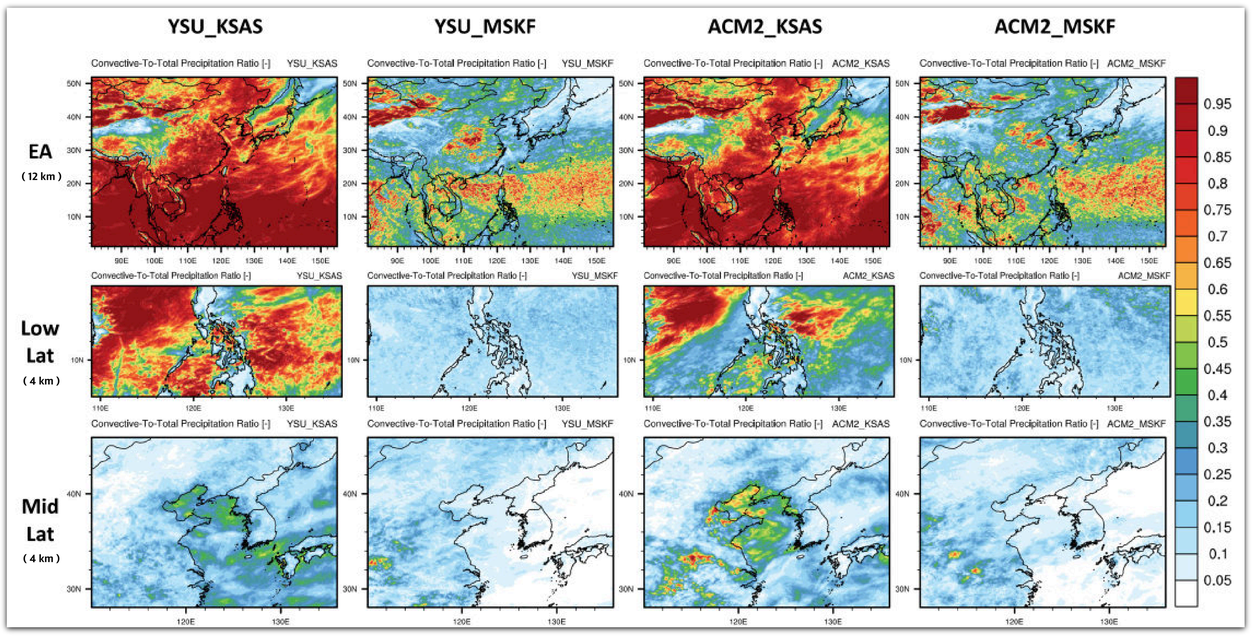
Environment

- **East Asia Summer Monsoon** is well appeared in 2022 Summer
- **Strong precipitation** over the **Southeast Asia** and **ITCZ over East Pacific**
- **Significant** precipitation over the **Philippines**
- **Moderate** precipitation in regions including parts of **Korea and Japan**

Model Characteristics

- **East Asia (Domain 01)**
  - All model show different spatial variation. KSAS overestimated
  - **ACM2_MSKF and YSU_MSKF show minimum bias**
  - **MSKF exhibit more localized** high precipitation areas compared to KSAS
- **Mid Latitude (Domain 03)**
  - All model has significant variations with underestimation.
  - MSKF shows a wider spread and higher intensity then KSAS
- **Low Latitude (Domain 02)**
  - KSAS shows higher precipitation intensities compared to IMERG, indicating potential underestimation in these models
  - MSKF display a more moderated and dispersed precipitation pattern, closer to IMERG but still with some variations
- **Overall**
  - **KSAS generally show higher precipitation biases** then MSKF

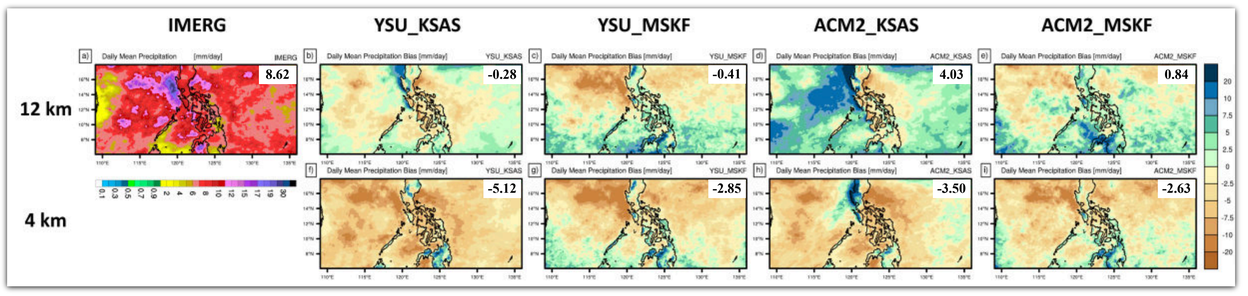
Convective Rainfall Ratio (CRR)



- **Low Latitude :** KSAS is well indicating the higher CRR values, as it is known (while MSKF is not)
- **Mid Latitude :** MSKF does not affected by PBL, but KSAS showing higher CRR with ACM2 than YSU
- **MSKF** show **consistent CRR value on different resolution**. Scale adaptive parameter well designed to show similar CRR on different resolution
- **KSAS** show **different CRR value on different resolution**. It is designed to reflect the change of CRR ration on different resolution

Resolution Effect

❖ Low Latitude

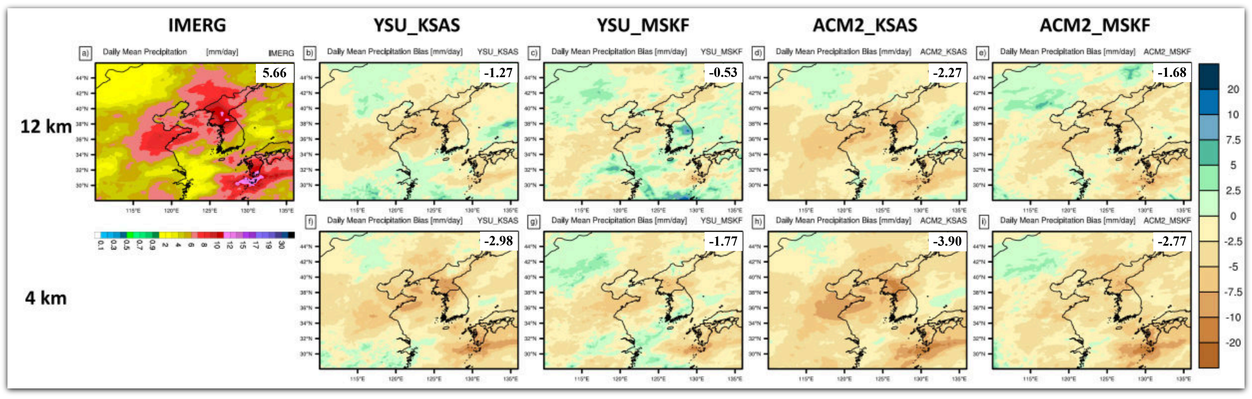


❖ **Model Characteristics**

	12 km	4 km
• YSU_KSAS	**lowest bias**	More underestimated, **larger negative bias than 12 km**
• YSU_MSKF	like YSU_KSAS, but with slightly increased bias	More underestimated, **larger negative bias than 12 km**
• ACM2_KSAS	**highest bias**, with significant overestimating	Significantly underestimated
• ACM2_MSKF	balanced than YSU_MSKF, but larger bias	More underestimated, **larger negative bias than 12 km**

- Overall, **MSKF** shows **larger negative bias on 4 km than 12 km**
- **KSAS** shows better estimation with **ACM2 on 4 km**, **YSU on 12 km**

❖ Mid Latitude



❖ **Model Characteristics**

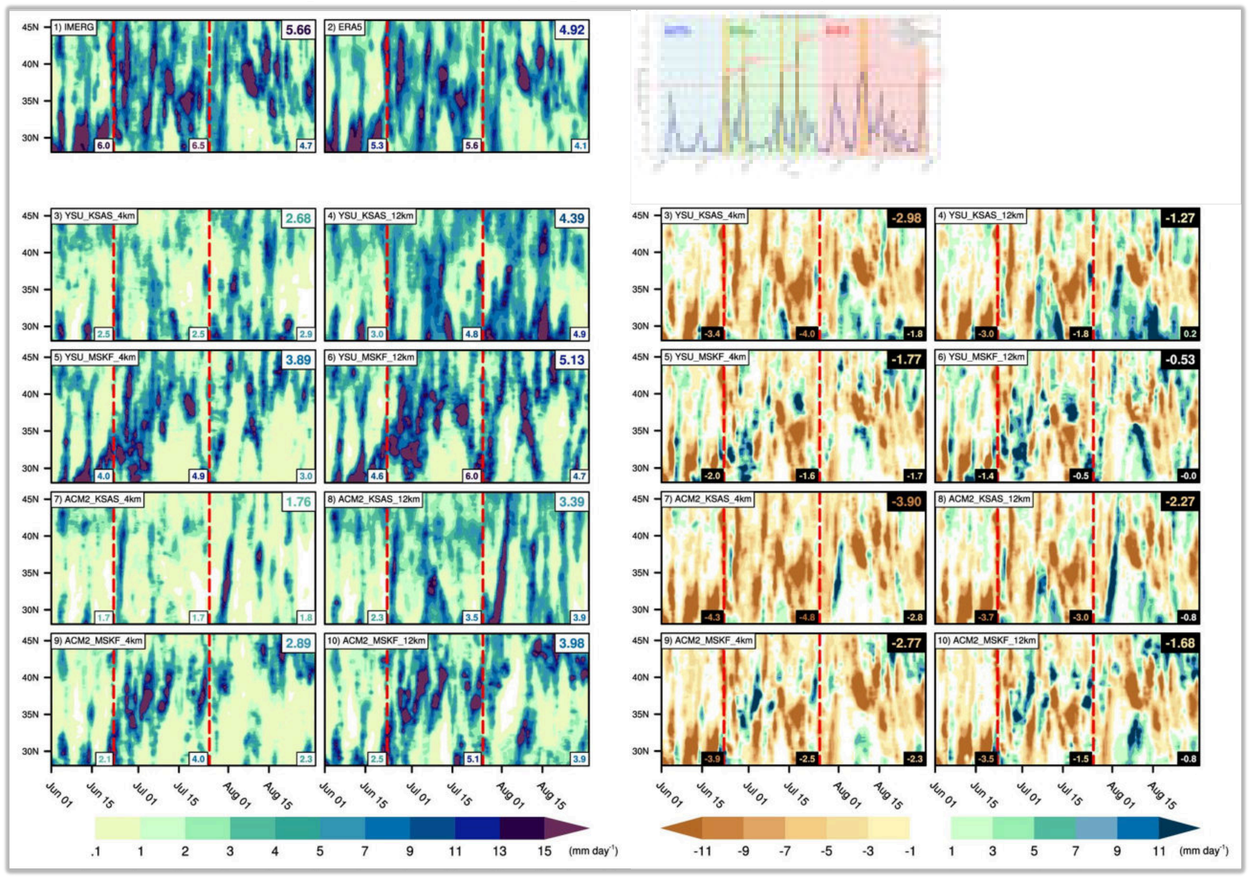
	12 km	4 km
• YSU_KSAS	moderate bias	**Bias is increased** compared to 12 km
• YSU_MSKF	**best estimation** among the 8 cases	**Low bias**, with improved performance on some regions
• ACM2_KSAS	larger negative bias, underestimating	**Significant large** negative bias
• ACM2_MSKF	reduced bias compared to ACM2_KSAS	**Larger negative bias with 4 km** than 12 km

- Higher resolution doesn't guarantee better precipitation predicting performance (Bias: 4 km > 12 km, KSAS > MSKF, ACM2 >YSU)
- **Every model underestimates** precipitation compared to IMERG
- **KSAS is more affected by resolution than MSKF**
- **4 km experiment is more affected by PBL** than 12 km experiment

Latitude & Resolution Sensitivity

- **KSAS performs better in low latitude** regions compared to mid latitude regions **at a 4 km resolution**.
- **MSKF** is not significantly affected by resolution in mid latitude regions but shows worse performance at a 4 km resolution in low latitude regions.
- **No consistent evidence that specific schemes consistently perform well predicting performance** regardless of variations in resolution, latitude, and combinations.

Time – Latitude Precipitation and Bias (compared to IMERG, only Domain 3) – Far East Asia



- IMERG and ERA5 precipitation are comparable, making both suitable for evaluation purposes.
- Most models underestimate precipitation, with **YSU_MSKF_12km (6)** showing better performance in capturing precipitation patterns.
- Significant underestimation is concentrated around mid-July, particularly in the **ACM2 (7-10)** schemes.
- Higher resolution **(3, 5, 7, 9)** does not consistently lead to improved performance.
- Precipitation patterns vary significantly across latitudes and over time.
- Most models fail to capture the high precipitation during the monsoon period.
- The models simulate a **late northward progression**, with precipitation lingering in southern latitudes longer than observed.

4. Summary

- **CPS with resolution**: CPS schemes are more influenced by resolution and latitude than by the combination of PBL
- **KSAS performance**: KSAS performs **better at 4 km resolution in low latitude regions** compared to mid latitude regions.
- **MSKF performance**: **less affected by resolution in mid latitude regions** but **worsens** at **4 km** in **low latitude** regions.
- **Inconsistency in scheme performance:** No CPS or PBL combination consistently performs well across all resolutions and latitudes, requiring careful selection for specific regions.
- **Precipitation Bias:** Most combinations underestimate precipitation, with higher resolutions not always improving performance and KSAS showing higher bias than MSKF.
- **PBL interaction with resolution:** The influence of PBL schemes becomes more affective at lower resolution (12 km).
- **Further Study**: Analysis with synoptic patterns related each scheme's properties.

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